

# Model Derivations

Companion notes to “Trade Tariffs and Exchange Rates”

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These notes derive three models with minimal exposition: the two-country benchmark, the two-layer nested three-country model with the reversal threshold, and the canonical competing-exporters model from the preferential-trade-agreement (PTA) literature.

## 1 Two-country model

### 1.1 Setup

Countries  $i \in \{A, B\}$ . Goods: tradable  $T_i$  (produced only in  $i$ , consumed in both) and non-tradable  $N_i$ . Labor is the only factor, immobile across countries, endowment  $L_i$ .

Technology (linear):

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i. \quad (1)$$

Perfect competition and the normalization  $P_{T_i} = 1$  (producer price of each tradable in its own currency) give

$$w_i = A_{T_i}, \quad P_{N_i} = \frac{w_i}{A_{N_i}} = \frac{A_{T_i}}{A_{N_i}}. \quad (2)$$

Let  $e$  be the nominal exchange rate: units of  $A$ 's currency per unit of  $B$ 's currency. The price of  $T_B$  in  $A$ 's currency is  $e$ ; the price of  $T_A$  in  $B$ 's currency is  $1/e$ .

Preferences (Cobb–Douglas, common weights):

$$U_i = C_{T_A i}^{\alpha_{T_A}} C_{T_B i}^{\alpha_{T_B}} C_{N_i}^{\alpha_{N_i}}, \quad \alpha_{T_A} + \alpha_{T_B} + \alpha_{N_i} = 1, \quad (3)$$

where  $C_{gi}$  is country  $i$ 's consumption of good  $g$ .

### 1.2 Demands

Cobb–Douglas expenditure shares are constant. With income  $I_i$  and consumer prices  $P_g^{(i)}$  (in  $i$ 's currency),

$$P_g^{(i)} C_{gi} = \alpha_g I_i \implies C_{gi} = \frac{\alpha_g I_i}{P_g^{(i)}}. \quad (4)$$

Under free trade  $I_i = w_i L_i$ ,  $P_{T_B}^{(A)} = e$ ,  $P_{T_A}^{(B)} = 1/e$ .

### 1.3 Equilibrium under free trade

Country  $A$ 's trade balance in  $A$ 's currency is export revenue minus import spending. From (4), the quantity  $B$  buys of  $T_A$  is  $C_{TAB} = \alpha_{T_A} I_B / (1/e) = \alpha_{T_A} w_B L_B e$ , valued at producer price 1 in  $A$ 's currency. The quantity  $A$  buys of  $T_B$  is  $C_{TBA} = \alpha_{T_B} w_A L_A / e$ , valued at price  $e$ . Hence

$$TB_A = 1 \cdot C_{TAB} - e C_{TBA} = \alpha_{T_A} w_B L_B e - \alpha_{T_B} w_A L_A. \quad (5)$$

Setting  $TB_A = 0$  (by Walras' law this is the only independent condition):

$$e = \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{w_A L_A}{w_B L_B} = \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{A_{T_A} L_A}{A_{T_B} L_B} \quad (6)$$

### 1.4 Tariff

Country  $A$  levies ad valorem tariff  $\tau \geq 0$  on imports of  $T_B$ . The consumer price in  $A$  becomes  $(1 + \tau)e$ ; producers in  $B$  still receive 1. Revenue is rebated lump-sum:

$$I_A = w_A L_A + T_A^{\text{rev}}, \quad T_A^{\text{rev}} = \tau e C_{TBA}. \quad (7)$$

Using  $C_{TBA} = \alpha_{T_B} I_A / [(1 + \tau)e]$ ,

$$I_A = w_A L_A + \frac{\tau}{1 + \tau} \alpha_{T_B} I_A \implies I_A = \frac{w_A L_A}{1 - \frac{\alpha_{T_B} \tau}{1 + \tau}}. \quad (8)$$

The trade balance (flows valued at producer prices; the tariff wedge is a domestic transfer):

$$TB_A = C_{TAB} - e C_{TBA} = \alpha_{T_A} w_B L_B e - \frac{\alpha_{T_B} I_A}{1 + \tau}. \quad (9)$$

Setting  $TB_A = 0$  and substituting (8):

$$e(\tau) = \frac{1}{1 + \tau} \cdot \frac{1}{1 - \frac{\alpha_{T_B} \tau}{1 + \tau}} \cdot \frac{\alpha_{T_B}}{\alpha_{T_A}} \cdot \frac{A_{T_A} L_A}{A_{T_B} L_B} \quad (10)$$

**Result 1.1.**  $e(\tau)$  is strictly decreasing in  $\tau$  for  $\alpha_{T_B} \in (0, 1)$ : a tariff appreciates the tariff-imposing country's currency.

*Proof.* Write  $e(\tau) = e(0) \cdot f(\tau)$  with  $f(\tau) = [(1 + \tau) - \alpha_{T_B} \tau]^{-1} = [1 + (1 - \alpha_{T_B})\tau]^{-1}$ . Since  $1 - \alpha_{T_B} > 0$ ,  $f' < 0$ .  $\square$

## 1.5 Relative wage, terms of trade, and real exchange rate

**Relative wage.** Wages in own currency are  $w_A = A_{T_A}$  and  $w_B = A_{T_B}$ , both constant. The relative wage in a common currency ( $A$ 's) is

$$\omega \equiv \frac{e w_B}{w_A} = \frac{A_{T_B}}{A_{T_A}} e \propto e. \quad (11)$$

A rise in  $e$  (depreciation of  $A$ 's currency) is a fall in  $A$ 's wage relative to  $B$ 's.

**Terms of trade.**  $A$ 's terms of trade are the price of its exports relative to its imports, in a common currency and at world (pre-tariff) prices:

$$\text{ToT}_A \equiv \frac{P_{T_A}}{e P_{T_B}} = \frac{1}{e}. \quad (12)$$

Combining (11) and (12): the exchange rate, the (inverse) common-currency relative wage, and the (inverse) terms of trade are the same object up to constants,

$$e \propto \frac{w_B e}{w_A} \propto \frac{1}{\text{ToT}_A}. \quad (13)$$

This holds because producer prices are unit labor costs ( $P_{T_i} = w_i/A_{T_i} = 1$ ): one factor and constant returns make the identification exact.

**Real exchange rate.** The consumer price index in each country is the Cobb–Douglas index over consumer prices (constants  $\kappa_i$  collect the share terms):

$$P_A = \kappa_A [P_{T_A}]^{\alpha_{T_A}} [(1 + \tau)e]^{\alpha_{T_B}} [P_{N_A}]^{\alpha_N} = \kappa_A [(1 + \tau)e]^{\alpha_{T_B}} \left(\frac{A_{T_A}}{A_{N_A}}\right)^{\alpha_N}, \quad (14)$$

$$P_B = \kappa_B [1/e]^{\alpha_{T_A}} [P_{T_B}]^{\alpha_{T_B}} [P_{N_B}]^{\alpha_N} = \kappa_B e^{-\alpha_{T_A}} \left(\frac{A_{T_B}}{A_{N_B}}\right)^{\alpha_N}. \quad (15)$$

The real exchange rate is

$$q \equiv \frac{e P_B}{P_A} = \text{const} \times e^{1 - \alpha_{T_A} - \alpha_{T_B}} (1 + \tau)^{-\alpha_{T_B}} = \text{const} \times e^{\alpha_N} (1 + \tau)^{-\alpha_{T_B}}, \quad (16)$$

so in logs

$$d \log q = \alpha_N d \log e - \alpha_{T_B} \frac{d\tau}{1 + \tau}. \quad (17)$$

Three observations. (i)  $q$  moves with  $e$ , attenuated by the non-tradable share  $\alpha_N$ . (ii) The tariff appreciates  $A$  in real terms directly, at any  $e$ , because it raises  $A$ 's consumer price index. (iii)  $P_{N_i} = A_{T_i}/A_{N_i}$  is a productivity ratio, so non-tradables affect the level of  $q$  but not its response to trade shocks except through the exponent  $\alpha_N$ .

## 1.6 The Marshall–Lerner condition

Let export volume depend on the price foreign buyers face,  $p^* = 1/e$ , and import volume on the price home buyers face,  $p = (1 + \tau)e$ . Write  $X(p^*)$  and  $M(p)$  with elasticities

$$\varepsilon_X \equiv -\frac{X'(p^*)p^*}{X} \geq 0, \quad \varepsilon_M \equiv -\frac{M'(p)p}{M} \geq 0. \quad (18)$$

The trade balance in  $A$ 's currency, at producer prices, is

$$\text{TB}_A(e, \tau) = X(1/e) - e M((1 + \tau)e). \quad (19)$$

**Response to the exchange rate.** Differentiate (19) with respect to  $e$  at  $\tau = 0$ :

$$\frac{\partial \text{TB}_A}{\partial e} = -\frac{X'(1/e)}{e^2} - M - e M'(e) = \frac{\varepsilon_X X}{e} - M + \varepsilon_M M, \quad (20)$$

using  $X' = -\varepsilon_X X e$  and  $e M' = -\varepsilon_M M$ . Evaluated at balanced trade ( $X = eM$ ):

$$\boxed{\frac{\partial \text{TB}_A}{\partial e} = M(\varepsilon_X + \varepsilon_M - 1)} \quad (21)$$

A depreciation improves the trade balance if and only if  $\varepsilon_X + \varepsilon_M > 1$ . This is the Marshall–Lerner condition. The “ $-1$ ” term is the valuation effect: a depreciation makes each unit of imports more expensive in home currency, worsening the balance one-for-one, and the two volume responses must jointly overcome it.

**Response to the tariff.** Differentiate (19) with respect to  $\tau$  at fixed  $e$ :

$$\frac{\partial \text{TB}_A}{\partial \tau} = -e M'((1 + \tau)e) e = \frac{\varepsilon_M}{1 + \tau} e M > 0 \quad \text{whenever } \varepsilon_M > 0. \quad (22)$$

A tariff improves the trade balance at a fixed exchange rate as long as import demand slopes down. (With a revenue rebate, income effects modify (22); in the Cobb–Douglas model of Section 1.4 the producer-price import value is  $\alpha_{TB} I_A / (1 + \tau) \propto [1 + (1 - \alpha_{TB})\tau]^{-1}$ , still decreasing in  $\tau$ .)

**Why Marshall–Lerner is necessary and sufficient for the conventional wisdom.** Equilibrium requires  $\text{TB}_A(e(\tau), \tau) = 0$ . By the implicit function theorem,

$$\frac{de}{d\tau} = -\frac{\partial \text{TB}_A / \partial \tau}{\partial \text{TB}_A / \partial e}. \quad (23)$$

The numerator is positive by (22): the tariff creates a surplus at the old exchange rate. To restore balance, the exchange rate must move in whichever direction *worsens* the trade balance. If Marshall–Lerner holds, that direction is appreciation ( $de < 0$ ): the conventional result. If Marshall–Lerner fails, depreciation worsens the balance instead, and the same tariff-induced surplus requires  $de > 0$ : the conventional result reverses. The sign of  $de/d\tau$  is the opposite of the sign of  $\varepsilon_X + \varepsilon_M - 1$ ,

which is why the condition is necessary and sufficient.

**The Cobb–Douglas case.** With unit expenditure shares, import volume is  $M = \alpha_{T_B} I/p$ , so  $\varepsilon_M = 1$ ; similarly  $\varepsilon_X = 1$ . Then  $\varepsilon_X + \varepsilon_M = 2 > 1$  and (21) is positive: Marshall–Lerner holds automatically, which is why the closed form (10) is monotone.

**Multilateral remark.** In the three-country model of Section 2, each bilateral relationship satisfies a Marshall–Lerner condition, and the tariff still improves  $A$ ’s overall balance at fixed rates. What the bilateral conditions do not determine is *how the required adjustment is split between  $e_{AB}$  and  $e_{AC}$* . That split depends on substitution across foreign origins ( $\rho$ ) relative to substitution between home and foreign goods ( $\eta$ ), which is the subject of Section 2.

## 2 Three-country nested model

### 2.1 Setup

Countries  $i \in \{A, B, C\}$ , each producing tradable  $T_i$  and non-tradable  $N_i$  with the same linear technology, wage equation, and normalization as (2). Exchange rates  $e_{ij}$  ( $i$ ’s currency per unit of  $j$ ’s),  $e_{ii} = 1$ , no-arbitrage  $e_{ij} = e_{ik}e_{kj}$ ; the free rates are  $e_{AB}, e_{AC}$ . Tariffs  $\tau_{ij} \geq 0$  ( $i$ ’s tariff on  $j$ ’s good). Consumer price of  $T_j$  in  $i$ :  $p_{ij} = (1 + \tau_{ij}) e_{ij}$ .

Preferences (two-layer nest):

$$C_i = C_{T_i}^{\alpha_T} C_{N_i}^{1-\alpha_T} \quad (\text{Cobb–Douglas outer}) \quad (24)$$

$$C_{T_i} = \left[ \alpha_D^{1/\eta} C_{T_i i}^{\frac{\eta-1}{\eta}} + (1 - \alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \quad (\text{home vs. imports, elasticity } \eta) \quad (25)$$

$$M_i = \left[ \sum_{j \neq i} \beta_j^{1/\rho} C_{T_j i}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \quad (\text{across origins, elasticity } \rho) \quad (26)$$

with home weight  $\alpha_D$ , origin weights  $\beta_j$  (symmetric case:  $\beta_j = 1/2$ ). The single-elasticity model is the special case  $\rho = \eta$ .

### 2.2 Price indices and expenditure shares

CES price indices (in  $i$ ’s currency):

$$P_{M_i} = \left[ \sum_{j \neq i} \beta_j p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \quad P_{T_i} = \left[ \alpha_D p_{ii}^{1-\eta} + (1 - \alpha_D) P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}. \quad (27)$$

Expenditure shares of total income  $I_i$  (rows sum to  $\alpha_T$ ):

$$s_{ii} = \alpha_T \alpha_D \left( \frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \quad (28)$$

$$s_{ij} = \alpha_T (1 - \alpha_D) \left( \frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_j \left( \frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \quad j \neq i. \quad (29)$$

### 2.3 Income and trade balances

Tariff revenue is rebated lump-sum. A fraction  $\tau_{ij}/(1 + \tau_{ij})$  of spending  $s_{ij}I_i$  is revenue, so

$$I_i = w_i L_i + \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij} I_i \implies I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij}}. \quad (30)$$

The shares depend only on prices, so (30) is a closed form.

Trade balances at producer prices (spending  $s_{ij}I_i$  is tariff-inclusive; divide by  $1 + \tau_{ij}$  to get the producer-price value):

$$\text{TB}_i = \sum_{k \neq i} \frac{s_{ki} I_k}{1 + \tau_{ki}} - \sum_{j \neq i} \frac{s_{ij} I_i}{1 + \tau_{ij}}, \quad (31)$$

with all quantities expressed in a common currency (this does not affect the zeros). Equilibrium:  $\text{TB}_B = \text{TB}_C = 0$ ;  $\text{TB}_A = 0$  follows by Walras' law.

### 2.4 Linearization at the symmetric point

Set  $L_i = 1$ ,  $A_{T_i} = A_{N_i} = 1$ ,  $\beta_j = 1/2$ , and evaluate at free trade ( $\tau_{ij} = 0$ ), where  $e_{AB} = e_{AC} = 1$  and all incomes equal 1. Baseline shares:

$$s_{ii} = \alpha_T \alpha_D \equiv a, \quad s_{ij} = \frac{1}{2} \alpha_T (1 - \alpha_D) \equiv \frac{m}{2}, \quad m \equiv \alpha_T (1 - \alpha_D). \quad (32)$$

Perturb by  $d\tau$  ( $A$ 's tariff on  $B$ ) and let  $\nu_j \equiv d \log e_{Aj}$  ( $\nu_A = 0$ ). Log price changes:  $d \log p_{ij} = \nu_j + \mathbf{1}\{(i, j) = (A, B)\} d\tau$ .

Index changes, from (27):

$$d \log P_{M_i} = \sum_{j \neq i} \beta_j d \log p_{ij}, \quad d \log P_{T_i} = \alpha_D d \log p_{ii} + (1 - \alpha_D) d \log P_{M_i}. \quad (33)$$

Explicitly:

$$d \log P_{M_A} = \frac{1}{2}(\nu_B + \nu_C + d\tau), \quad d \log P_{M_B} = \frac{1}{2}\nu_C, \quad d \log P_{M_C} = \frac{1}{2}\nu_B. \quad (34)$$

Share changes, from (28)–(29):

$$d \log s_{ii} = (1 - \eta)(d \log p_{ii} - d \log P_{T_i}), \quad (35)$$

$$d \log s_{ij} = (1 - \rho)(d \log p_{ij} - d \log P_{M_i}) + (1 - \eta)(d \log P_{M_i} - d \log P_{T_i}). \quad (36)$$

Income changes, from (30) (the revenue term is first order for country  $A$  only):

$$d \log I_A = \frac{m}{2} d\tau, \quad d \log I_B = \nu_B, \quad d \log I_C = \nu_C. \quad (37)$$

Differentiate (31) for  $i = B, C$  at the baseline. Each term  $sI/(1 + \tau)$  contributes  $\frac{m}{2}(d \log s +$

$d \log I - d\tau$ ), where  $d\tau = d\tau$  appears only in the  $A$ -to- $B$  flow:

$$dTB_B = \frac{m}{2} \left[ (d \log s_{AB} + \frac{m}{2} d\tau - d\tau) + (d \log s_{CB} + \nu_C) - (d \log s_{BA} + \nu_B) - (d \log s_{BC} + \nu_B) \right] = 0, \quad (38)$$

$$dTB_C = \frac{m}{2} \left[ (d \log s_{AC} + \frac{m}{2} d\tau) + (d \log s_{BC} + \nu_B) - (d \log s_{CA} + \nu_C) - (d \log s_{CB} + \nu_C) \right] = 0. \quad (39)$$

Substituting (35)–(36) and collecting terms in  $(\nu_B, \nu_C, d\tau)$  gives a linear system. Define

$$D \equiv 3\alpha_D(\eta - 1) + \rho + 1, \quad \rho^* \equiv 3[1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D)]. \quad (40)$$

The system evaluates to (each entry verified symbolically):

$$-\frac{mD}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \frac{m}{4} \begin{pmatrix} \rho + \rho^*/3 \\ -\rho + \rho^*/3 \end{pmatrix} d\tau. \quad (41)$$

Invert using  $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ :

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = -\frac{1}{3D} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \rho + \rho^*/3 \\ -\rho + \rho^*/3 \end{pmatrix} d\tau = \frac{1}{3D} \begin{pmatrix} -(\rho + \rho^*) \\ \rho - \rho^* \end{pmatrix} d\tau. \quad (42)$$

**Result 2.1** (Threshold). *At the symmetric point,*

$$\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0} = -\frac{\rho + \rho^*}{3D} < 0, \quad \boxed{\left. \frac{d \log e_{AC}}{d\tau} \right|_{\tau=0} = \frac{\rho - \rho^*}{3D}} \quad (43)$$

with  $D > 0$  for  $\eta \geq 1$  (local stability).  $A$  appreciates against  $B$  for all parameters;  $A$  depreciates against the untariffed  $C$  if and only if  $\rho > \rho^*$ .

## 2.5 Impossibility under a single elasticity

Impose  $\rho = \eta = \sigma$  (the single-elasticity CES). Then

$$\rho^* - \sigma = 3 + 3\alpha_D(\sigma - 1) - 3\alpha_T(1 - \alpha_D) - \sigma = \sigma(3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T). \quad (44)$$

**Result 2.2** (Impossibility). *If  $\alpha_D \geq 1/3$ , both terms in (44) are nonnegative and the second is positive, so  $\rho^* > \sigma$  for every  $\sigma > 0$ : the conventional appreciation against  $C$  survives at any elasticity. At  $\alpha_D = 1/3$ ,  $\alpha_T = 3/4$  (the symmetric baseline calibration  $\alpha_{T_j} = \alpha_N = 1/4$ ):  $\rho^* - \sigma = 1/2$  and  $D = 2\sigma$ , so*

$$\left. \frac{d \log e_{AC}}{d\tau} \right|_{\tau=0} = \frac{\sigma - \rho^*}{3D} = -\frac{1}{12\sigma} \rightarrow 0^-. \quad (45)$$

## 2.6 Nominal versus real exchange rate in the three-country model

The quantities of Section 1.5 carry over:  $e_{Aj}$  is proportional to  $j$ 's relative wage in  $A$ 's currency and to the inverse of  $A$ 's bilateral terms of trade with  $j$ . The real rate against  $C$  is  $q_{AC} = e_{AC}P_C/P_A$  with  $P_i = \kappa_i (P_{T_i}^{\text{own}})^{\alpha_T} (P_{N_i})^{1-\alpha_T}$ , where  $P_{T_i}^{\text{own}}$  is the tariff-inclusive tradable index (27) in  $i$ 's own currency and  $P_{N_i} = A_{T_i}/A_{N_i}$  is constant. Hence

$$d \log q_{AC} = d \log e_{AC} + \alpha_T (d \log P_{T_C}^{\text{own}} - d \log P_{T_A}^{\text{own}}). \quad (46)$$

At the symmetric point, converting the indices to own currency ( $d \log P_{T_C}^{\text{own}} = d \log P_{T_C} - \nu_C$ ):

$$d \log P_{T_A}^{\text{own}} = (1 - \alpha_D) \frac{1}{2} (\nu_B + \nu_C + d\tau), \quad (47)$$

$$d \log P_{T_C}^{\text{own}} = \alpha_D \nu_C + (1 - \alpha_D) \frac{1}{2} \nu_B - \nu_C = (1 - \alpha_D) (\frac{1}{2} \nu_B - \nu_C). \quad (48)$$

Substituting and using  $m = \alpha_T(1 - \alpha_D)$ :

$$d \log q_{AC} = \nu_C + m \left( \frac{1}{2} \nu_B - \nu_C - \frac{1}{2} \nu_B - \frac{1}{2} \nu_C - \frac{1}{2} d\tau \right) = \left( 1 - \frac{3m}{2} \right) \nu_C - \frac{m}{2} d\tau. \quad (49)$$

As in (17), the tariff appreciates  $A$  in real terms directly through the price index (the  $-\frac{m}{2}d\tau$  term), and the pass-through from the nominal rate is attenuated. Setting (49) to zero with  $\nu_C = \frac{\rho - \rho^*}{3D} d\tau$  gives the real-rate reversal threshold:  $(2 - 3m)(\rho - \rho^*) = 3mD$ , and since  $D = 3\alpha_D(\eta - 1) + \rho + 1$  this is linear in  $\rho$ , with solution

$$\boxed{\rho_q^* = \frac{(2 - 3m) \rho^* + 3m[3\alpha_D(\eta - 1) + 1]}{2 - 6m}} \quad (\rho_q^* > \rho^* \text{ for } m \in (0, \frac{1}{3})). \quad (50)$$

At the benchmark ( $\alpha_D = 0.8$ ,  $\alpha_T = 0.4$ ,  $\eta = 1.5$ , so  $m = 0.08$ ):  $\rho^* = 3.96$  and  $\rho_q^* = 4.93$ . Real depreciation against  $C$  requires more diversion than nominal depreciation, because part of the nominal movement is absorbed by the tariff's direct effect on  $A$ 's price index.

## 3 Canonical PTA model: competing exporters

### 3.1 Setup

Countries  $A$  (importer),  $B$  and  $C$  (exporters). Two goods:  $x$  and numeraire  $y$ . Endowments:  $A$  holds only  $y$ ;  $B$  and  $C$  hold  $x$  (and possibly  $y$ ). Quasilinear preferences in each country,

$$U_i = u_i(x_i) + y_i, \quad u_i' > 0, \quad u_i'' < 0. \quad (51)$$

$A$  levies specific tariffs  $t_B, t_C$  on imports of  $x$  from  $B$  and  $C$ . Let  $q_j$  be the price received by exporter  $j$  (the world price of its good) and  $p$  the domestic price of  $x$  in  $A$ . Both origins sell in  $A$  at the same domestic price:

$$p = q_B + t_B = q_C + t_C. \quad (52)$$



### 3.2 Demands and market clearing

From  $u'_i(x_i) = (\text{local price of } x)$ :

$$M_A(p) = x_A^d(p), \quad E_j(q_j) = \omega_j - x_j^d(q_j), \quad j = B, C, \quad (53)$$

with  $M'_A < 0$  and  $E'_j > 0$ . Market clearing for  $x$ :

$$M_A(p) = E_B(q_B) + E_C(q_C) = E_B(p - t_B) + E_C(p - t_C). \quad (54)$$

Equation (54) determines  $p$  given  $(t_B, t_C)$ ; then  $q_B, q_C$  follow from (52).

### 3.3 Comparative statics of a discriminatory tariff

Totally differentiate (54) with respect to  $t_B$ , holding  $t_C$ :

$$M'_A \frac{dp}{dt_B} = E'_B \left( \frac{dp}{dt_B} - 1 \right) + E'_C \frac{dp}{dt_B} \implies \frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0, 1). \quad (55)$$

Therefore

$$\frac{dq_B}{dt_B} = \frac{dp}{dt_B} - 1 = \frac{E'_C - M'_A}{E'_B + E'_C - M'_A} \cdot (-1) < 0, \quad (56)$$

$$\frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \quad (57)$$

**Result 3.1** (Viner–Mundell). *A discriminatory tariff on B worsens B's terms of trade and improves C's terms of trade. Equivalently, a preferential tariff cut toward B (lower  $t_B$ ) improves B's and worsens C's terms of trade.*

### 3.4 Mapping to the exchange-rate model

In the model of Section 2,  $C$ 's terms of trade improving corresponds to  $C$ 's wage rising in common-currency units, i.e.  $e_{AC}$  rising:  $A$  depreciates against  $C$ . The canonical PTA model therefore always delivers the reversal. The reason is visible in (54):  $A$  has no domestic production of  $x$ , so tariffed demand can only switch between  $B$  and  $C$ . There is no home–import margin. In the notation of Section 2, the PTA model sets the home share to zero, and (44) with  $\alpha_D = 0$  gives  $\rho^* - \sigma = 3(1 - \alpha_T) - \sigma$ , which is negative for moderate  $\sigma$ : the impossibility disappears when there is nothing to switch home to. The nested three-country model adds the home margin  $(\eta, \alpha_D)$  and shows that with realistic home bias the diversion force must clear the threshold  $\rho^*$  before the PTA logic controls the exchange rate.